MIL-A-27198B(USAF)
29 December 1987
SUPERSEDING
MIL-A-27198A(USAF)
27 FEBRUARY 1961

# MILITARY SPECIFICATION

## ALTIMETER, PRESSURE AAU-7/A

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the requirements for one type of precision pressure, -1000 +80,000 feet, integrally lighted altimeter, designated.

1.2 Classification. Altimeters shall be of the following type, as specified (see 6.2):

AAU-7/A
AAU-7A/A Antireflective coating on cover glass, improved reliability, and TSO-C10b requirements.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-B-601 | Boxes, Wood, Cleated Plywood |
| PPP-B-636 | Box, Shipping, Fiberboard |

Military

| | |
|---|---|
| MIL-P-116 | Preservation, Methods Of |
| MIL-C-675 | Coating Of Glass Optical Elements (Antireflection) |
| MIL-C-5541 | Chemical Films For Aluminum and Aluminum Alloys |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6610
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| | |
|---|---|
| MIL-P-7105 | Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT |
| MIL-S-7742 | Screw Threads, Standard, Optimum Selected |
| MIL-A-8625 | Anodic Coatings, For Aluminum And Aluminum Alloys |
| MIL-L-25467 | Lighting, Integral, Instrument, General Specification For |
| MIL-L-27160 | Lighting, Instrument, Integral, White, General Specification For |
| MIL-D-70327 | Drawings, Engineering And Associated Lists |
| MIL-C-83488 | Coating, Aluminum, Ion Vapor Deposited |

STANDARDS

Federal

| | |
|---|---|
| FED-STD-595 | Colors |

Military

| | |
|---|---|
| MIL-STD-129 | Marking For Shipment And Storage |
| MIL-STD-130 | Identification Marking of U.S. Military Property |
| MIL-STD-143 | Specification And Standards Order Of Precedence For The Selection |
| MIL-STD-794 | Parts and Equipment, Procedures For Packaging And Packing Of |
| MIL-STD-810C | Environmental Tests Methods And Engineering Guidelines |
| MIL-STD-889 | Dissimilar Metals |
| MS28105 | Window, Dial-Aircraft Instrument Cover, Glass |
| MS33549 | Case, Instrument, 2-3/4 Dial, With Sump, Standard Dimensions For |
| MS33558 | Numerals And Letters, Aircraft Instrument Dial, Standard Form Of |
| MS33585 | Pointers, Dial, Standard Design Of Aircraft Instrument |

(Copies of specifications, standards, drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial



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(Applications for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other information services.)

3. REQUIREMENTS

"3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2 and 6.4)."

3.2 Components parts. The altimeter shall consist of a case, bezel, dial, pointers, and operating mechanism.

3.3 Selection of specification and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-143 except as provided in 3.3.1 and 3.3.2.

3.3.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case commercial utility parts, like screws, bolts, nuts, and cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.3.2 Standard parts. With the exceptions in 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Materials

3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the contracting activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.4.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts except where magnetic materials are essential.

3.4.3 Nonferrous materials. Nonferrous materials shall be used for all parts of the altimeter except where ferrous materials are essential.

3.4.3.1 Recycled and reclaimed materials. Recycled and reclaimed materials shall be used to the maximum extent possible without jeopardizing the end use of the item.



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3.4.4 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to be met in storage or normal service.

3.4.5 Protective treatment. When materials are used in the construction of the altimeter that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.5 Design and construction. The altimeter shall conform to figure 1. The design and movement shall be of the aneroid type. The design shall be reasonably simple to permit overhaul or repair without requiring numerous special tools and fixtures. The altimeter shall be so constructed as to withstand, without failure, the normal strains, jars, vibrations, and such other conditions as are incident to shipping storage installation, and service.

3.5.1 Mechanism adjustments. The mechanism shall be provided with means of adjusting or correcting the indications to conform to the performance required herein, in case this becomes necessary during the life of the altimeter. The calibration adjustment shall not affect the temperature performance of the altimeter. This means of adjustment shall be simple to manipulate and of such nature that adjustment can be accomplished by tools ordinarily possessed by an instrument repairman.

3.5.2 Zero setting system. A zero setting system, controlled by a setting knob, shall be provided and shall permit the altimeter to be set to indicate zero altitude at any existing ground level pressure throughout the range from 3.10 to 28.1 (inclusive) inches Hg. The digits 31.0 through 28.1, in 0.1-inch increments, shall be provided on a subdial barometric scale graduated to read 0.02 inch Hg, as shown on figure 2. This scale shall be so located that it will not be obscured by the rotating disk. Positive stops shall be provided at or near the extremes of the barometric scale so that the scale cannot misset.

3.5.2.1 Adjustment of setting. The zero setting system shall be provided with a means for adjusting or correcting the relative setting of the altitude pointers and the zero setting system indications during the service like of the altimeter. When installed on the instrument board, this adjustment shall be accessible from the face of the altimeter.

3.5.3 TSO-C10b (AAU-7A/A only). The altimeter shall meet the requirements of Federal Aviation Agency TSO-C-10B (see 6.5).

3.6 Performance. The altimeter shall be capable of satisfactory operation when subjected to the following conditions:

a. A scale reading not to exceed 25 feet difference from the "correction difference" over the range specified herein.



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b. Case leakage that will not cause the altimeter reading to change more than 100 feet in 1 minute at approximately 40,000 feet.

c. Scale errors not to exceed the specified tolerances when tested at room temperature, $-35^{\circ}$, $+45^{\circ}$, $-54^{\circ}$, and $+71^{\circ}$C.

d. Errors not to exceed 20 feet difference from the reading in the normal operating position.

e. Errors not to exceed those specified for the altitude at which the altimeter is tested.

f. Temperature ranging from $-54^{\circ}$ to $+71^{\circ}$C, and after prolonged exposure at $-62^{\circ}$C.

g. Errors not to exceed 20 feet when vibrated at an amplitude of between 0.003 and 0.005 inch at frequencies from 500 to 3,000 Hz.

h. Relative humidity up to 100 percent.

i. Exposure to fungus growth as will be encountered in tropical climates.

j. Exposure to simulated salt sea atmosphere.

k. Exposure to sand and dust particles as will be encountered in desert areas.

l. Exposure to simulated rainfall.

m. Acceleration forces of 20g for 30 seconds.

n. A deflection not exceed $1^{\circ}$ when held 5 inches from the center of an aircraft compass.

o. When subjected to 2,000 applications of 50,000 altitude, 3 cycles of thermal shock and 30 days storage.

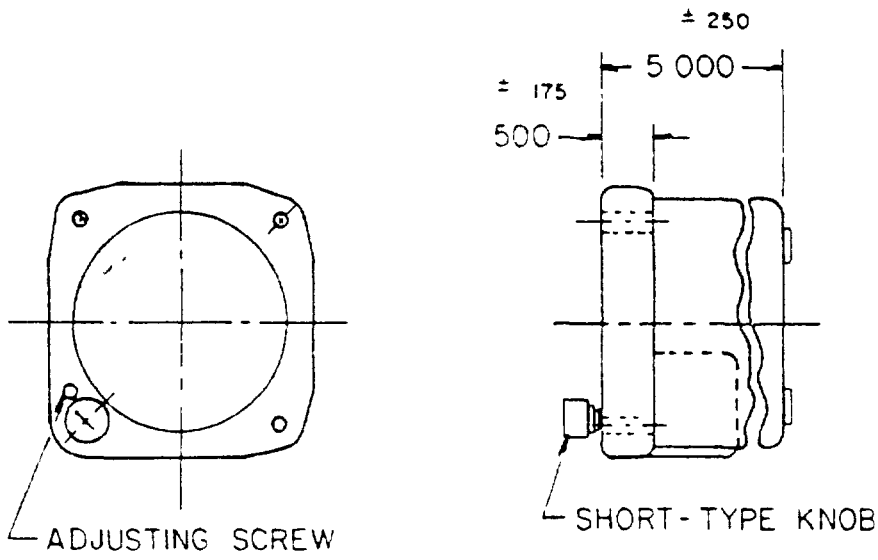
p. (For AAU-7A/A only). 3,175 applications of an obsolete pressure variation equivalent to an altitude of approximately 50,000 feet, three applications of thermal shock, and 30 days of storage. The endurance (reliability) requirement for 3,175 applications on the three altimeters (see 4.4.1.1), will demonstrate a minimum NTBF of 1,400 hours based on the hi-squared distribution at 90 percent confidence with no failures allow.

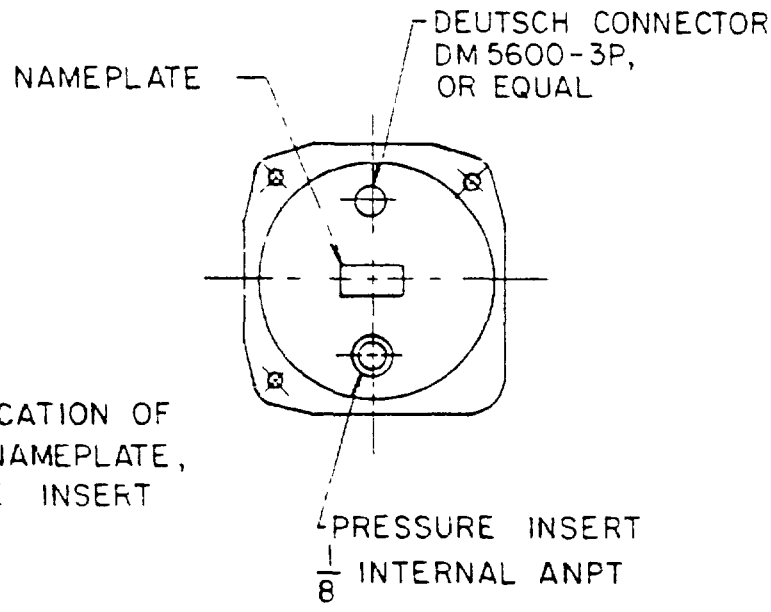
3.7 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-70327 shall govern changes in the manufacturer's part numbers.

3.8 Case. The case shall consist of a body and bezel ring and shall conform to figure 1. The design of the case shall be subjected to the approval of the contracting activity.



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RELATIVE LOCATION OF CONNECTOR, NAMEPLATE, AND PRESSURE INSERT OPTIONAL.

CASE SHALL BE IN ACCORDANCE WITH STANDARD MS33549, EXCEPT AS SHOWN

DIMENSION IN INCHES

FIGURE 1 Case.



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3.8.1 Body. The body shall be made of nonferrous, low-density metal, shall be uniform in tenture, and shall have a smooth surface with durable black finish. The mounting flange portion of the body shall have a dull black finish.

3.8.1.1 Reinforcement. In the event the construction of the body is of aluminum alloy, the insert shall be anodically treated, except for the threads. The inside of the body around the insert shall be strengthened by a reinforcement to prevent damage to the body when the connection is tightened during the installation of the altimeter.

3.8.1.2 Bezel. The bezel shall be made of nonferrous low-density metal and shall have a durable dull black finish. The bezel shall be held in place by means of screws.

3.8.1.3 Cover glass. The cover glass of the altimeter shall conform to MS28105. It shall be properly sealed, and means shall be provided for replacement of the cover glass by removal of the bezel. The bezel shall be held in place by means of screws properly secured by lockwashers or similar devices of approved design. If the cover glass is used as part of the lighting system, its size and shape may deviate from those dimensions specified in MS28105 to the extent required for proper lighting design. If the cover glass and bezel form an integral assembly they may be replaced as a unit.

3.8.1.3.1 Distance from dial to cover glass. The distance from dial to cover glass shall be small as practicable and shall not exceed 0.250 inch.

3.8.1.3.2 Reflection surfaces (AAU-7A/A) only. All reflecting glass surfaces shall be provided with a reflection reducing coating which meets the requirements of MIL-C-675, in addition to withstanding the environmental conditions specified herein, except that the following reflectance tolerances shall apply.

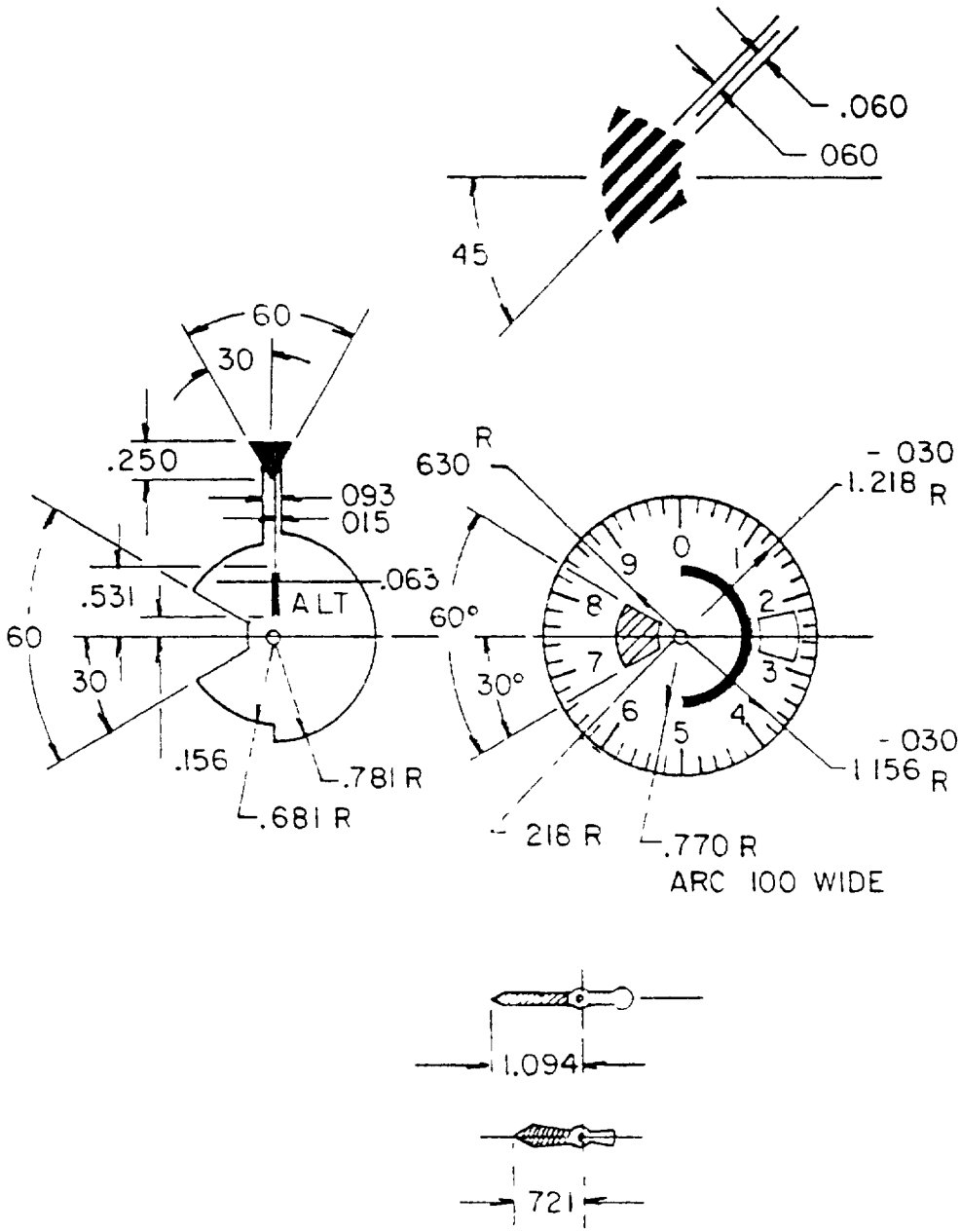
| Angle of Incidence | Wavelength Millimicrons | Percent Reflectance |
|---|---|---|
| $0^{\circ}$ | 450 to 675 | 0.6 maximum |
| $0^{\circ}$ | 425 to 700 | 0.5 average |
| $30^{\circ}$ | 450 to 625 | 1.0 maximum |
| $30^{\circ}$ | 425 to 700 | 0.5 average |

3.8.1.4 Shipping plug. The static pressure connection shall be provided with a vented plug of acceptable material and design. Each plug shall incorporate a vent consistent of a hole 0.062 $\pm$0.016 inch in diameter.

3.9 Pointers and disk. Two pointers and a disk with pointer extension shall be provided. These shall be concentrically mounted graduated uniformly as shown on figure 2, as follows: A large pointer conforming essentially to MS33585-3, except as shown on figure 2, and cross section as approved by the contracting activity, indicating hundreds of feet and making 1 revolution for each 1,000 feet of altitude; an intermediate pointer conforming to MS33585-4, indicating thousands of feet and making 1 revolution for each 10,000 feet of altitude and a round disk as shown on figure 2 with a pointer extension indicating tens of thousands of feet and making 1 revolution for each 100,000 feet.



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DIMENSIONS IN INCHES UNLESS
OTHERWISE SPECIFIED
TOLERANCES

DECIMALS ± .005
ANGLES ± 0° 15'

FIGURE 2 Dial



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3.9.1 Pointer adjustment. The pointers shall be free to swing to either side of their zero positions without engaging mechanical stop and shall be firmly attached to the mechanism, yet readily adjusted.

3.10 Integral lighting. The indicator shall be integrally lighted with 2 parallel lighting circuits, a white lighting circuit and a red lighting circuit, as shown on figure 3. The red lighting shall be in accordance with MIL-L-25467 and the white lighting shall be in accordance with MIL-I-27160. In the event there is any conflict between this specification and either MIL-L-25467 or MIL-L-27160, this specification shall govern. Although resistors shall not be permitted in the original design of lighting systems, space shall be allocated for incorporation of a trimming resistor in each lighting circuit within the indicator. The magnitude of the trimming resistor shall be specified by the contracting activity in terms of the brightness decrease required. The lighting shall be subjected to approval by the contracting activity.

3.10.1 Lighting. The integral lighting shall be comparable to and in brightness balance with a "prime standard" indicator. The prime standard shall be an indicator of the type described in this specification. The prime standard shall be an indicator of the type described in this specification. The prime standard shall have applied to its lighting circuits the specific voltage which the contracting activity has established as being necessary to assure equality of presentation with respect to the lighting with adjacent but dissimilar instruments. The contractor shall submit the test procedure for determining compliance with the paragraph to the contracting activity for approval.

3.11 Dial.

3.11.1 Markings. All markings shall be durable to withstand the usage encountered in service. The style and proportions of numerals and letters laced on the dial shall conform to MS33558.

3.11.2 Numerals. Numerals shall distinctly indicate the graduation to which each applies. If practicable, each numeral shall be so placed that the center of mass of the numeral is on the radial line joining the appropriate graduations and the center of the dial. Any confusion resulting in doubt as to which graduation the numeral applies shall be cause for rejection. When several numerals are used in one group, the space between the numerals shall be approximately 0.016 inch.

3.11.3 Finish and dimensions. The following markings shall be finished in color No. 37875 of FED-STD-595. The dimensions shall be as shown in the following tabulation



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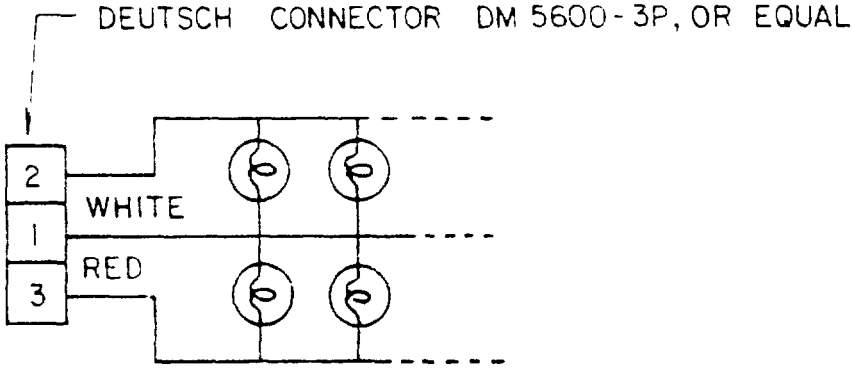



FIGURE 3 Wiring



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| Markings | Height or Length Inch ±0.015 | Width of Line or Graduation Inch ±0.005 |
|---|---|---|
| Numerals 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9 | 0.312 | 0.029 |
| 100 foot graduations | 0.156 min | 0.030 |
| Zero setting pressure scale and index | 0.125 | 0.015 |
| 20 foot graduations | 0.093 | 0.015 |
| Lettering AIT | 0.188 | 0.023 |
| Lettering 100 FEET | 0.046 | 0.008 |
| Shaded portions of pointer and disk | ----- | ----- |
| Extension line from 10,000 feet pointer to triangular indicator | ----- | 0.015 |
| Shaded warning area on face (45° angle with horizontal) | ----- | 0.060 |

3.11.3.1 Markings. The following markings shall be finished in durable dull black color No. 37038 of FED-STD-595. The dimensions shall be as shown in the following tabulation:

| Markings | Height or Length Inch ±0.015 | Width of Line or Graduation Inch ±0.005 |
|---|---|---|
| Lettering 80,000 FT | 0.046 | 0.015 |
| Lettering AAU-7/A | 0.063 | 0.015 |
| Unshaded portions of pointer and background of dial and disk | ----- | ------- |

3.11.4 Visibility of dial. The pointer, numerals, at least 0.062 inch of the shortest graduations, and other specified markings on the dial shall be visible from any point within the frustrum of a cone whose side makes an angle of 20° with a perpendicular to the dial, and whose small diameter is the aperture of the instrument case.

3.12 Threads.

3.12.1 Screw. Unless otherwise specified, the threads of all machine screws, 0.060 inch or larger in diameter, shall conform to MIL-S-7742.

3.12.2 Pipe. Pipe threads shall conform to MIL-P-7105.

3.13 Weight. The weight of the altimeter completely assembled and without connecting fittings and mounting screws shall not exceed 1.85 pounds.



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3.14 Finishes and protective coatings.

3.14.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625 except as follows:

a. Dials, small holes, and case inserts need not be anodize.

b. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

c. Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

d. Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

e. When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.14.2 Steel parts . Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be avoided when satisfactory alternative processes can be used.

3.15 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.16 Workmanship. The altimeter shall be constructed and finished in a thoroughly workmanlike manner in accordance with MIL-STD-454, requirement 9. Particular attention shall be given to neatness and throughness of soldering, wiring, marking of parts and assemblies, plating, painting, riveting, machine screw assembly, welding, brazing, and freedom of parts from burrs and sharp edges.

3.16.1 Dimensions. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances affect the Interchangeability, operation, or performance of the altimeter, they shall be held or limited accordingly.

3.16.2 Gears. Gear assemblies shall be properly aligned meshed, and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustments, gear assemblies shall be free from backlash.

3.16.3 Cleaning. The altimeter shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material, after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.



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3.17 Scale error card. An altimeter scale error card, Form AFTO 139 shall be furnished by the contractor with each altimeter. The card shall be filled in waterproof ink with errors found in final inspection.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of know defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection and of the altimeter shall be classified as follows.

a. First article inspection (see 4.4).

b. Acceptance inspection (see 4.5).

4.3 Inspection condition. Unless otherwise specified all inspections shall be performed in accordance with the test condition specified in 4.3.1 of this specification.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified, the test shall be conducted at ambient atmospheric pressure and at a temperature of 25 $\pm 5^{\circ}$C.

4.3.2 Vibration stand. Whenever a vibration stand is specified, it shall vibrate at any desired frequency between 300 and 3,000 Hz and shall subject the altimeter to vibration so that a point on the instrument case will describe in a plans inclined $45^{\circ}$ to the horizontal plane, a circle of not less than 0.003 nor more than 0.005 inch diameter, or a circle of not less than 0.018 nor more than 0.020 inch diameter as specified in the tests.

4.3.3 Removing friction. Unless otherwise specified, the altimeter shall be tested in its normal operation position and shall be vibrated to remove the friction before a test reading is taken. Vibration shall be applied by means of electromechanical vibrators, (Edwards Vibrators or equivalent), or a vibration stand if specified. Tapping the instruments for the purpose of removing the friction is unacceptable. The amount of vibration applied shall always be sufficient to remove all of the friction from the instrument.



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4.3.4 Standard pressures. The standard pressures used in calibrating the altimeters shall be as specified in table I.

4.3.5 Adjustments. No adjustments shall be made to the altimeter after initiation of acceptance or first article tests.

4.3.6 Barometer. During acceptance or first article tests a Hass Type A-1 barometer or equivalent, equipped with a Type PSR-1A Photo Scanner, or equivalent, proper calibrated for accuracy shall be used to measured scale error, hystersis and after effect. The barometer shall be properly maintained and checked in accordance with the handbook or instructions furnished by the barometer manufacturer.

4.3.7 Test sequence. The test shall be conducted in the same sequence as listed in 4.6 test methods, unless a different sequence is approved by the contracting activity.

4.4 First article inspection. When 3 first article test samples are specified they shall be subjected to all the tests specified herein.

4.4.1 First article inspection samples. The contractor shall subject 3 or 6 altimeters to the first article inspection specified in 4.4.

4.4.1.1 First article test schedule. When 3 first article test samples are specified they shall be subjected to all tests specified herein, when 6 first article test samples are specified, the samples shall be grouped and subjected schedule

| Group I<br>3 Altimeters | Group II<br>3 Altimeters |
|---|---|
| Sampling Plan A (4.5.2.1) | "Endurance, Stability, and |
| Environmental (4.6.14) | Shelf Life (4.6.18)" |
| Magnetic Property (4.6.15) | |



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TABLE I. Scale Error.

| Standard Altitudes Feet | Equivalent Atmospheric Pressure Mercury | | Scale error tolerances (+Feet) | |
|---|---|---|---|---|
| | mm | in | Room temperature -35°C and +45°C | -54° and +71°C |
| 0 | 760.00 | 29.921 | 30 | 60 |
| 500 | 746.37 | 29.385 | 35 | |
| 1,000 | 732.93 | 28.856 | 35 | |
| 1,500 | 719.70 | 28.335 | 40 | |
| 2,000 | 706.65 | 27.821 | 40 | 80 |
| 2,500 | 693.80 | 27,315 | 45 | |
| 3,000 | 681.14 | 26.817 | 45 | |
| 4,000 | 656.38 | 25.842 | 50 | |
| 5,000 | 632.36 | 24.896 | 55 | |
| 6,000 | 609.05 | 23,978 | 60 | |
| 8,000 | 564.51 | 22.224 | 70 | |
| 10,000 | 522.65 | 20.577 | 80 | 160 |
| 12,000 | 483.34 | 19.029 | 90 | |
| 14,000 | 446.46 | 17,577 | 100 | |
| 15,000 | 428.90 | 16.886 | 105 | 210 |
| 16,000 | 411.90 | 16.216 | 110 | |
| 18,000 | 379.53 | 14,942 | 120 | |
| 20,000 | 349.25 | 13,750 | 130 | 260 |
| 22,000 | 320.96 | 12,636 | 140 | |
| 25,000 | 282.03 | 11.103 | 155 | 310 |
| 30,000 | 225.69 | 8.885 | 180 | 360 |
| 35,000 | 178.83 | 7.041 | 205 | |
| 40,000 | 140.66 | 5.538 | 230 | 460 |
| 45,000 | 110.62 | 4.335 | 255 | |
| 50,000 | 86.99 | 3.425 | 280 | 560 |
| 55,000 | 68.40 | 2.693 | 600 | |
| 60,000 | 53.79 | 2.118 | 800 | 1,600 |
| 70,000 | 33.26 | 1.3096 | 1,200 | |
| 80,000 | 20.57 | 0.8099 | 1,500 (Room Temp)<br>2,250 (-35° + +45°C) | |
| | | | | 3,000 |

4.4.2 First article inspection samples for contracting activity. The contractor shall submit first article samples to the acquisition activity.

4.5 Acceptance inspection. Acceptance inspection shall consist of:

a. Individual tests

b. Sampling plans and tests



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4.5.1 Individual tests. Each altimeter shall be subjected to the following tests as described under 4.6 TEST METHODS:

a. Examination of product

b. Zero setting scale

c. Case leakage

d. Scale error at room temperature

e. Hysteresis

f. After effect

g. Position error

h. Friction

i. Scale error at -35°C

j. Lighting

k. Scale error at +45°C

4.5.2 Sampling plans and tests.

4.5.2.1 Sampling plan A. One altimeter selected at random from each 100 or less produced on the contract or order shall be subjected to the following tests as described under 4.6 TEST METHODS:

a. Individual tests (see 4.5.1)

b. Scale error at -54°C

c. Scale error at +71°C

d. Vibration error

4.5.2.2 Sampling plan B. Unless otherwise specified, 3 or 6 altimeters (see 4.4.1) selected at random from the first 15 items of the contract or order shall be subjected to the following tests as described under 4.6 TEST METHODS:

a. Sampling plan A

b. High temperature exposure

c. Low temperature exposure

d. Humidity

e. Fungus

f. Salt fog



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g. Vibration error

h. Dust (fine sand)

i. Rain

j. Acceleration

k. Magnetic property

l. Endurance, stability, and shelf life

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no item still on hand or later produced shall be accepted until the extent and cause of failure are determined and appropriately corrected. The contractor shall explain to the Government representatives the cause of failure and the action taken to preclude recurrence. After corrections have been made, all necessary tests shall be repeated.

4.5.2.3.1 Individual tests may continue. For operational reasons, individual tests or other sampling plans may continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the contracting activity of all defects likely found and methods of correcting them.

4.6 Test methods.

4.6.1 Examination of product. Each altimeter shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.6.2 Zero-setting scale. The altimeter shall be subjected to atmospheric pressure. The pressure scale shall be set successively (ranging from 28.1 to 31.0 as specified in table II). The altimeter shall be mounted on the vibration stand and vibrated simultaneously and the pointer indication noted. After the series of readings has been recorded, the difference between the reading at the setting of 29.92 and each other setting shall be found. These differences shall be recorded and compared with the values in the correct differences column. The recorded values shall not differ from the correct differences by more than 25 feet at any setting.

4.6.3 Case leakage. A vacuum sufficient to produce a change of reading of the pointers approximately 40,000 feet shall be applied slowly to the static pressure connection of the altimeter, at which point connection tubing shall be printed off or otherwise completely sealed. The change in reading in 1 minute shall not exceed 100 feet.

4.6.4 Scale error at room temperature. The altimeter shall be tested for scale errors as follows: The barometric setting scale shall be set to 29.92. While the



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altimeter is subjected to atmospheric pressure, the reading of the pointers shall be taken and the error of this indication determined by comparison with a barometer. This error shall be used later in the test specified in 4.6.6. Without changing the barometric setting, the altimeter shall be subjected successively to the pressures specified in table I. The decrease or increase in pressure shall not exceed a rate of 3,000 fpm. The altimeter shall remain at the pressure corresponding to each test point for at lease 1 minute, but not more than 5 minutes before a tests reading is taken. The scale errors shall not exceed the tolerances specified in table I. Without vibration being applied to the altimeter the movement of the pointer shall be observed for any irregular motion. From 0 to 25,000 feet, the pointer shall not stop and jump more than 120 feet. From 25,000 to 50,000 feet the pointer shall not stop and jump more than 250 feet and above 50,000 feet the pointer shall not stop and jump more than 850 feet. The scale error test shall be conducted with the altimeter in an altitude or vacuum chamber.

TABLE II. Zero-Setting Mechansim.

| Setting Pressure Scale | Correct Differences |
|---|---|
| 28.1 | 1,727 |
| 28.5 | 1,340 |
| 29.0 | 863 |
| 29.5 | 392 |
| 29.92 | 0 |
| 30.5 | 531 |
| 30.90 | 893 |
| 31.0 | 983 |

4.6.5 Hysteresis. Not more than 15 minutes after the altimeter has been first subjected to the pressure corresponding to the upper limit of the scale in the test specified in 4.6.4, the pressure shall be increased at a rate corresponding to a decrease in altitude of approximately 3,000 fpm until the pressure corresponding the first test point given in table III is reached. The altimeter shall remain at this pressure for at least 5 minutes, but not more than 15 minutes, before the test reading is taken. After the reading has been taken, the pressure shall be further increased at the specified rate until the pressure corresponding to the second test point given in table III is reached. The altimeter shall remain at this pressure for at least 1 minute, but not more than 10 minutes before the test reading is taken. After the reading has been taken, the pressure shall be further increased at the specified rate until atmospheric pressure is reached. The reading of the altimeter at either of the 2 test points shall not differ from the reading of the altimeter for the corresponding altitude in the tests specified in 4.6.4 by more than the tolerance given in table III.

4.6.6 After effect. Not less than 1 minute and not more than 5 minutes after the completion of the test specified in 4.6.5, the pointers shall have returned to their original reading corrected for any change in atmosphere pressure within the tolerance given in table III as the third test point.



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TABLE III. Hystersis.

| Test Points | Standard Altitude | Equivalent Atmospheric Pressure | Permissible Variation |
|---|---|---|---|
| First point | 25,000 | 281.9 | 100 |
| Second point | 20,000 | 349.1 | 100 |
| Third point | ------ | Ground level | 50 |

4.6.7 Position error. The altimeter reading shall be taken while it is in a vertical position (dial up) and after it is vibrated. The altimeter shall then be rotated 90° to its normal operating position. Readings will be taken, after vibration, in the 12, 3, 6, and 9 o'clock positions. None of the readings shall differ from the 12 o'clock reading by more than 20 feet. This test shall be conducted with the altimeter mounted in a test fixture, which with a fixed base can be used to rotate the altimeter to each desired position. The test fixture shall be mounted on a Edwards vibrator, or equivalent, of the type specified in 4.4.3 for the purpose of removing the friction. The vibrator shall be strapped to the instrument case and remain in this position for the duration of this position error test. The amount of vibration applies shall be sufficient to remove all of the friction from the instrument.

4.6.8 Friction. The friction test may be combined with the scale error at room temperature test. The altimeter shall be tested for friction at the test points specified in table IV. Vibration shall be provided by means of tapping or by use of electromechanical vibrations mounted on the test panel. The instrument shall be subjected to a constant rate of decrease of pressure not to exceed 3,000 fpm until the pressure reaches the equivalent of 500 ±100 feet shall be decreased to a maximum of 600 fpm. Rate of pressure increases or decreases will be measured by means of a rate of climb calibrated for accuracy. The pressure shall be decreased at the rate specified above to the pressure equivalent of the altitude of the desired test point, then held constant while 2 readings are taken, the first before the instrument is vibrated and the second after vibration is applied. The difference of any 2 such readings shall be recorded as friction and shall not exceed tolerance of table IV.

4.6.9 Scale error -35°C. For a period of not less than 12 hours prior to this test, the altimeter shall not have been subjected to any test involving operation at other atmospheric pressure. The test specified in 4.6.4 shall be repeated, except that the temperature of the altimeter during the test shall, in this case, be approximately -35°C maintained for 3 hours. The test points shall be those given in table I. The scale errors of this test shall not exceed the tolerances specified in table I.

4.6.10 Lighting. Each lighting circuit shall be properly connected individually and tested to show compliance with requirements specified in 3.10.

4.6.11 Scale error at -54°C. The test specified in 4.6.9 shall be repeated with the difference that the temperature of the altimeter shall be approximately -54°C instead -35°C maintained for 3 hours. The scale errors of this test shall not exceed the tolerance specified in table I. During and after this test, the setting system shall operate properly.



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TABLE IV. Friction Error Tolerances.

| Altitude (feet) | Tolerance (feet +) |
|---|---|
| 1,000 | 70 |
| 2,000 | 70 |
| 3,000 | 70 |
| 5,000 | 70 |
| 10,000 | 80 |
| 15,000 | 90 |
| 20,000 | 100 |
| 25,000 | 120 |
| 30,000 | 140 |
| 35,000 | 160 |
| 40,000 | 180 |
| 50,000 | 250 |
| 60,000 | 450 |
| 70,000 | 650 |
| 80,000 | 850 |

4.6.12 Scale error at +71°C. For a period of not less than 12 hours prior to this test, the altimeter shall not have been subjected to any test involving operation at other than atmospheric pressures. The altimeter shall be subjected to a temperature of +71°C for 3 hours. The altimeter shall then be subjected to the tests specified in 4.6.4. The scale errors of this test shall not exceed the tolerances specified in table I. After this test the setting system shall operate properly.

4.6.13 Vibration error. The altimeter shall be mounted on a vibration stand in its normal operation and secured by flange screws as it would be on an aircraft panel. The altimeter shall be subjected to vibration in accordance with 4.3.2. The natural frequency of the altimeter shall not occur during vibration. While the altimeter is being vibrated, the previous amplitude of pointer shall not exceed 20 feet, and the pointer vibration from its original position shall not exceed 20 feet.

4.6.14 Environmental. The altimeter shall be subjected to the following tests, conducted in accordance with the specified Method of MIL-STD-810C. Upon completion of each test, the altimeter shall be subjected to and meet the requirements specified in 4.6.4.

4.6.14.1 High temperature exposure. Method 502.1, procedure I the temperature shall be $+71^{\circ} \pm 2^{\circ}$C.

4.6.14.2 Low temperature exposure. Procedure II.

4.6.14.3 Humidity. Method 507.1, procedure I except for a period of 10 cycles. The altimeter shall be thoroughly examined in not less than 48 hours after completion of the humidity exposure, and there shall be no evidence of internal or external corrosion and deterioration that would affect subsequent operation of the altimeter.



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4.6.14.4 Fungus. Procedure I. A the end of the test period, the altimeter shall be thoroughly examined to ascertain that no fungus growth has occurred.

4.6.14.5 Salt fog. Method 509.1, procedure I for 50 hours. There shall be no corrosion or other damage as a result of this test which will affect subsequent operation of the altimeter.

4.6.14.6 Vibration. Method 514.2, procedure IA.

4.6.14.7 Dust (fine sand). Method 510.1, procedure I.

4.6.14.8 Rain. Method 506.1, procedure II.

4.6.14.9 Acceleration. Method 513.2, procedure I.

4.6.15 Magnetic property. The magnetic property of the altimeter shall be determined in terms of the deflection of a free magnet, approximately 1.50 inches long, in a magnetic field with a horizontal intensity of 0.18 ±0.001 oersted when the altimeter is held in various positions of an east-west line 5 inches from the center of the magnet. An aircraft compass, with compensation magnets removed therefrom, may be used as the free magnet for this test. The maximum deflection of the magnet shall not exceed 1°.

4.6.16 Scale error at +45°C. For a period of not less than 12 hours prior to this test, the altimeter shall to have been subjected to any test involving operation at other than atmospheric pressures. The test specified in 4.6.4 shall be repeated, except that the temperature of the altimeter during the test shall, in this case, be +45° ±5°, -0°C maintained for 3 hours. The test points shall be those given in table I. The scale errors of this test shall not exceed the tolerances specified in table I.

4.6.17 Lighting. Each lighting circuit shall be properly connected individually and tested to show compliance with the requirements specified in 3.10.

4.6.18 Endurance, stability, and shelf life (AAU-7/A). The altimeter shall be subjected to and shall meet the requirements of the individual tests. The room temperature scale error readings will be referred to later in this test as the original scale error readings. The instrument shall be subjected to a pressure cycling process consisting of 2,000 applications of an absolute pressure variation equivalent to approximately 50,000 feet pressure altitude. After 500, 1,000 and 1,500 cycles the altimeter shall be subjected to 2 cycles of temperature shock Method 503.1 of MIL-STD-810C except that the high temperature shall be +71°C in lieu of +85°C. Each pressure cycle shall consist of decreasing the pressure at a rate of 6,000 ±2,000 fpm until 50,000 ±5,000 feet is reached. The pressure shall be maintained at this pressure altitude for not less than 2 minutes. The pressure shall then be increased at a rate of 6,000 ±2,000 fpm until ambient pressure altitude is reached. After a minimum of 2 minutes at ambient pressure the next cycle shall be commenced. This is to be continued until 2,000 cycles have been completed after which the altimeter shall be rested at ambient pressure for approximately 48 hours. The contractor may remove the altimeter at any desired time during the cycling process to determined the altimeter performance at that time. Following completion of all cycling and the 48-hour rest period the altimeter may be reset by means of the zero setting adjustment mechanism. The amount of reset shall be limited to 100 feet. The instruments shall then be subjected to and meet the requirements for the individual tests.



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In addition, the room temperature scale errors shall not deviate from the original scale error readings by more than the values specified in table V.

TABLE V. Scale Errors Variation.

| Standard Altitude Feet | Scale Error Variation Tolerances Feet |
|---|---|
| 0 | 30 |
| 500 | 30 |
| 1,000 | 30 |
| 1,500 | 35 |
| 2,000 | 35 |
| 2,500 | 35 |
| 3,000 | 35 |
| 4,000 | 40 |
| 5,000 | 45 |
| 6,000 | 45 |
| 8,000 | 50 |
| 10,000 | 55 |
| 12,000 | 60 |
| 14,000 | 65 |
| 15,000 | 70 |
| 16,000 | 70 |
| 18,000 | 75 |
| 20,000 | 80 |
| 22,000 | 85 |
| 25,000 | 90 |
| 30,000 | 105 |
| 35,000 | 120 |
| 40,000 | 130 |
| 45,000 | 140 |
| 50,000 | 155 |
| 55,000 | 180 |
| 60,000 | 405 |
| 70,000 | 655 |
| 80,000 | 900 |

4.6.18.1 Storage and inspection. Following completion of the tests specified in 4.6.18 the altimeter shall be placed in storage for a period of not less than 30 days. During the 30-day period the instrument shall remain as dormant as practical. The mechanism shall not be exercised other than that caused by normal ambient barometric pressure variations. The temperature shall remain essentially constant, however, normal room temperature variations are permissible. After the 30-day period, the altimeter shall again be subjected to and shall meet the requirements of the individual tests. In addition to meeting the tolerances specified in table I for scale error, the readings shall not vary from the original scale error readings by more than the values specified in table V.



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4.6.18.2 Endurance, stability, and shelf line (AAU-7A/A only). The altimeter shall be subjected to and shall meet the individual tests. The room temperature scale error readings will be referred to hereafter as the original scale error readings. The altimeter shall be subjected to a pressure variation equivalent to a pressure altitude of approximately 50,000 feet. After 500, 1,000 and 1,500 cycles, the altimeter shall be subjected to two cycles of temperature shock in accordance with Method 503.1 of MIL-STD-810C except that the high temperature shall by 71°C in lieu of 85°C. Each pressure cycle shall consist of decreasing the pressure at a rate of 6,000 ± 2,000 fpm until 50,000 ± 5,000 feet is reached. The pressure shall be maintained at this pressure altitude for not less than 2 minutes. The pressure shall be increased at a rate of 6,000 ± 2,000 fpm until ambient pressure altitude is reached. This procedure shall be continued until 3,175 cycles have been completed after which the altimeter shall be rested at ambient pressure for approximately 48 hours. The contractor may check the altimeter performance at any time during the cycling process. Following the completion of the cycling and the 48-hour rest period, the altimeter may be reset by means of the zero setting adjustment mechanism. After reset, the altimeter shall again meet the individual tests. The room temperature scale errors shall not deviate from the original scale error readings by more than the values specified in table V.

4.7 Inspection of packaging. The inspection of the preservation and interior marking shall be in accordance with group A and B quality conformance inspection requirements, section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Altimeter shall be cleaned in accordance with process C-1, of MIL-P-116.

5.1.1.2 Drying. Altimeter shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each altimeter shall be packaged in quantity unit packs of one each in accordance with Method IIC of MIL-P-116. Each altimeter shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the altimeter when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each altimeter shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.



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5.1.3 Industrial. The Industrial preservation of altimeter shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Altimeter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Altimeter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during damage domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable rules and regulations.

5.2.4 Industrial. The packaged altimeter shall be packed in accordance with ASTM D3951.

5.2 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The AAU-7/A altimeter covered by this specification is intended for use on aircraft to indicate the height of the aircraft above some reference point, assuming standard conditions of temperature and pressure.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether sampling plan B tests are to be conducted.

c. Level of packaging and packing desired.

d. Type of altimeter required (see 1.2).

6.3 Definitions.

6.3.1 Scale error. Scale error is the difference between the correct value of pressure altitude corresponding to the absolute pressure and the indications of the altimeter.



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6.3.2 Amplitude. Amplitude is the extent of vibratory movement as measured from one extreme position to the opposite extreme position.

6.4 First article. A first article is required for inspection and approval (see 3.1 and 4.4).

6.5 TSO-C10B. The TSO application procedure is presented in Regulations of the Administrator dated 1 July 1962, subchapter C, subpart A, paragraph 514.2-TSO authorization. It will be the responsibility of the contractor to submit data to the FAA and obtain FAA approval is obtained and a copy of such approval will be submitted to the contracting officer.

6.6 Subject term (key word) listing.

AAU-7/A
AAU-7A/A
Altimeter
Integrally Lighted
Pressure

6.7 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Preparing Activity:
AIR FORCE -71

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2 DOCUMENT TITLE |
|---|---|
| MIL-A-27198B | ALTIMETER, PRESSURE AAU-7/A |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4 TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)* _____ |

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

| 7a NAME OF SUBMITTER *(Last, First, MI)* – Optional | b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8 DATE OF SUBMISSION *(YYMMDD)* |

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